Darwin Core Data Package (DwC-DP)

31 October 2026

Overview

The building blocks

1. Darwin Core List of Terms

Darwin Core terms provide **identifiers, labels, definitions, and usage guidance for concepts** used to describe taxa, organisms, occurrences, events, material entities, identifications, locations, geological context, measurements and assertions, agents, media, protocols, references, and related biodiversity information.

https://dwc.tdwg.org/list/

The building blocks

2. Darwin Core Conceptual Model (DwC-CM)

The Darwin Core Conceptual Model (DwC-CM) provides a **high-level framework that shows how Darwin Core (DwC) classes relate to one another** in typical biodiversity information workflows. Darwin Core formally defines a set of terms grouped by classes; DwC-CM clarifies the conceptual meaning of those classes and the relationships among them so implementers can make consistent, interoperable design choices across different technologies.

https://dwc.tdwg.org/cm/

The building blocks

3. Frictionless Data Package

Frictionless Data Package is an open, **lightweight container format** and specification used to bundle a collection of data files together with standardized metadata and schemas.

https://specs.frictionlessdata.io/

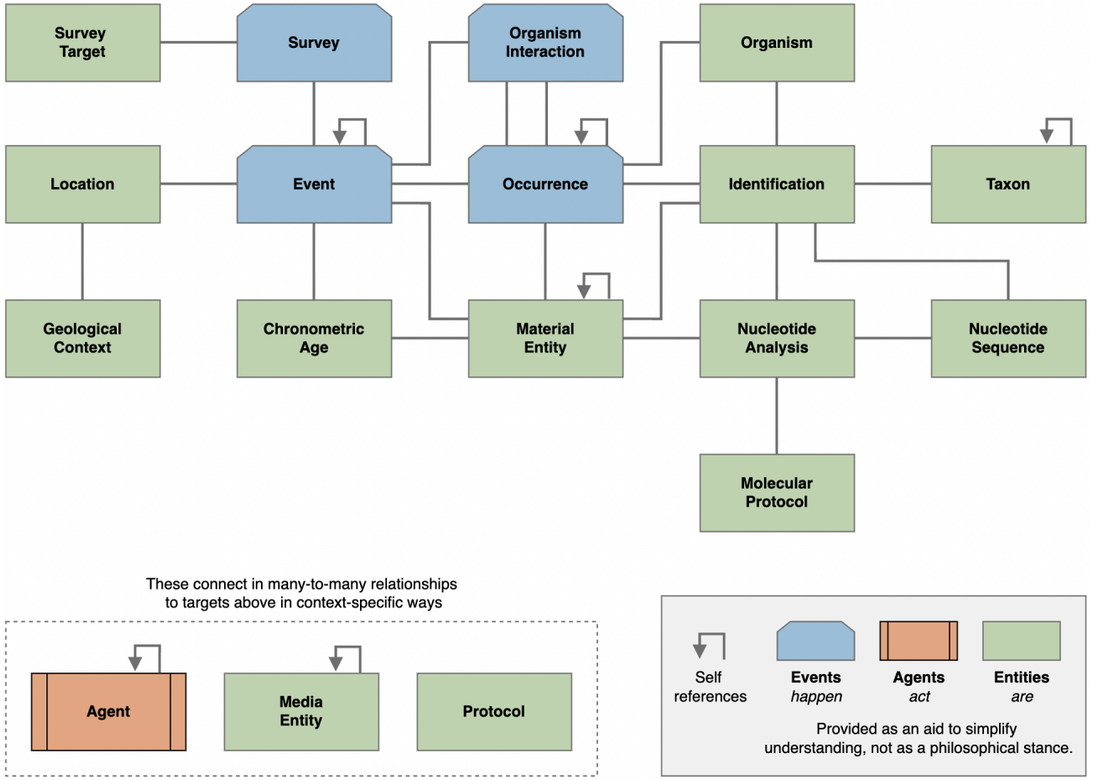
The building blocks

4. Darwin Core Data Package (DwC-DP)

Darwin Core Data Package (hereafter referred to as DwC-DP) is a **community-developed container format** to exchange biodiversity data. It extends the Data Package specification (developed by Frictionless Data) as an implementation for the Darwin Core Conceptual Model. This document specifies the requirements for a dataset to comply with DwC-DP.

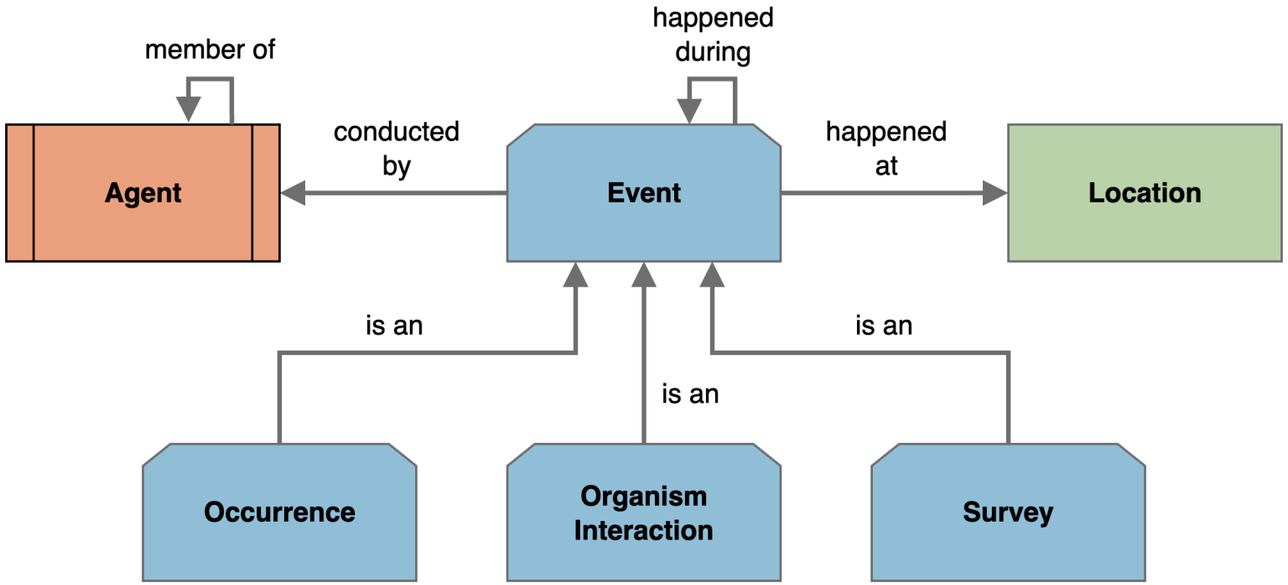
https://dwc.tdwg.org/dp/

Darwin Core Conceptual Model



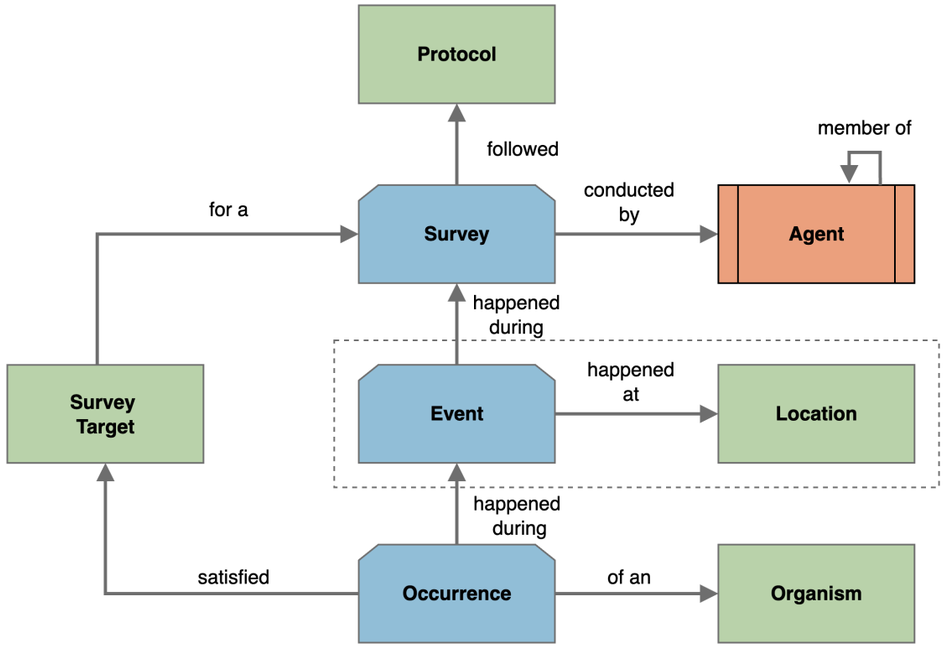
Darwin Core Conceptual Model

In Darwin Core, an **Event** is an action, a process, or a set of circumstances occurring at some place during some interval of time.



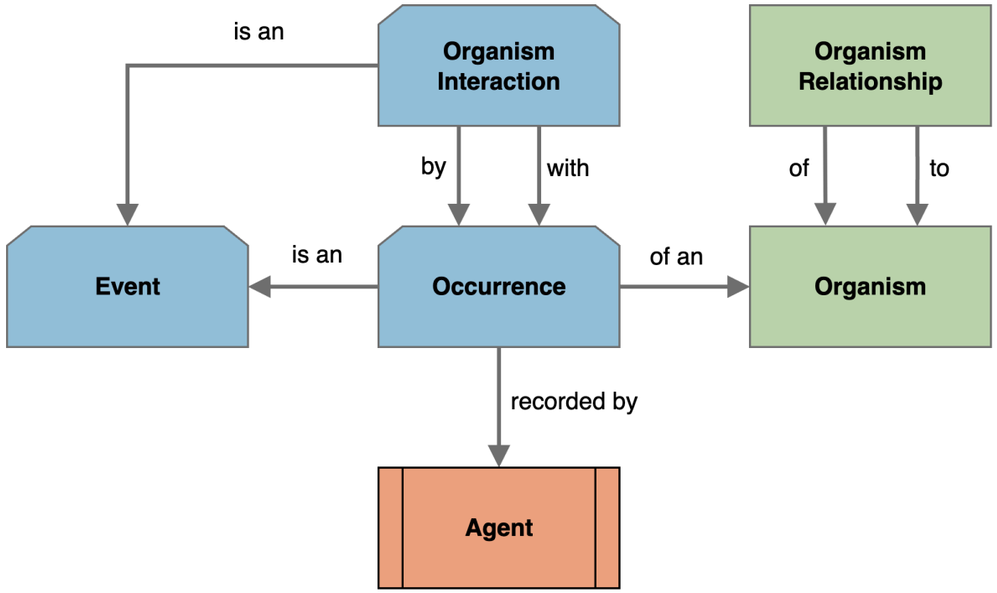
Darwin Core Conceptual Model

In Darwin Core, a **Survey** refers to an Event that is a biotic survey or inventory, which, with sufficiently detailed information, can support not only evidence of presence of Organisms but also absences of detection and estimations of abundance.



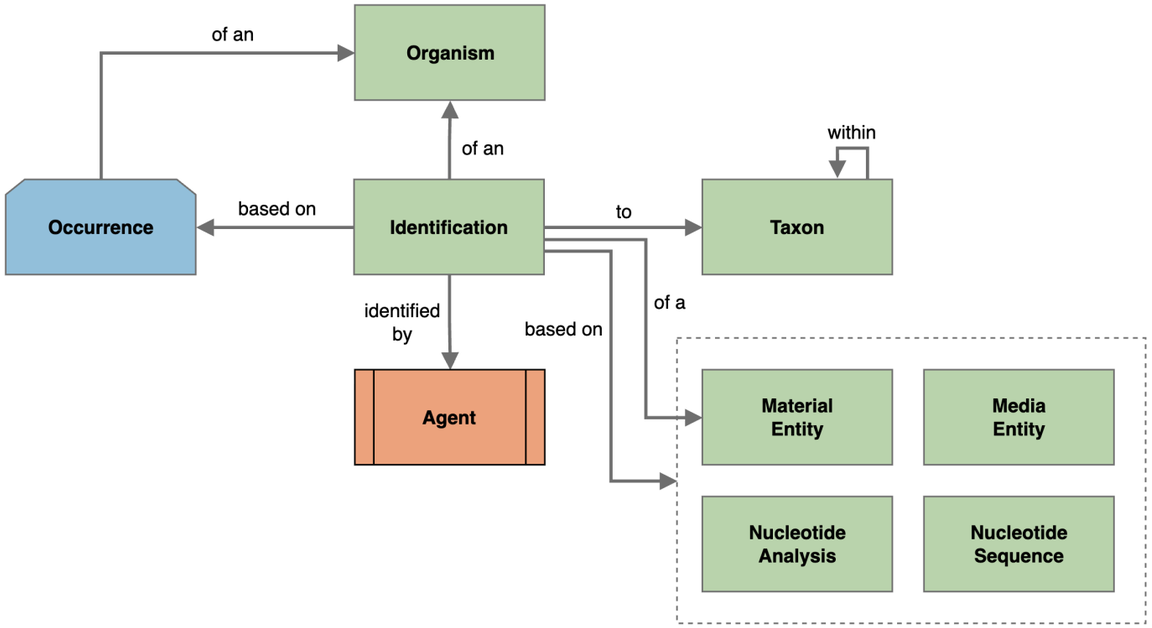
Darwin Core Conceptual Model

An **Occurrence** is an Event that establishes the state of an Organism at a particular place and time. The state of an Organism covered by the Occurrence class consists of attributes of an Organism that can differ between Occurrences, such as lifeStage, sex, etc.



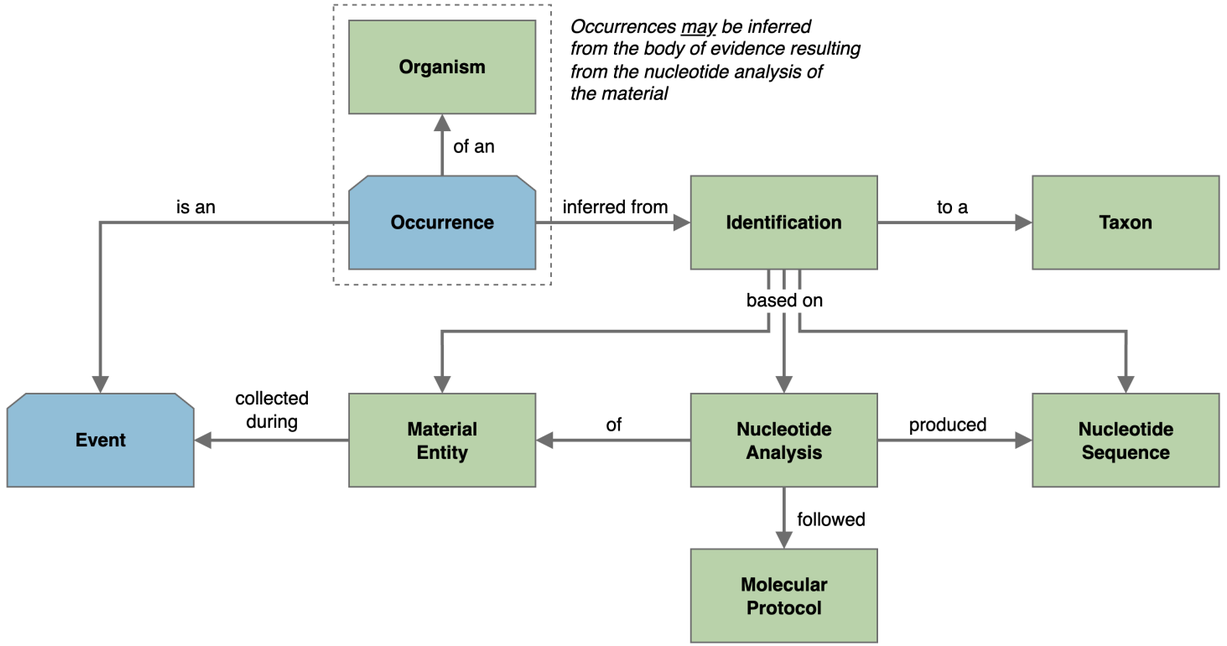
Darwin Core Conceptual Model

In Darwin Core, an **Identification** is defined as “A classification of a resource according to a classification scheme”. For Organisms, this means a taxonomic determination (i.e., the assignment of a Taxon to an Organism).



Darwin Core Conceptual Model

A **Nucleotide Analysis** is a link between a **Nucleotide Sequence** or an **Identification**, and an Event and a Material Entity from which it was derived, using a specified **Molecular Protocol**.



Darwin Core Data Package

DwC-DP content

- EML metadata file `eml.xml` (optional)
- JSON descriptor `datapackage.json`
- At least one resource file representing a data package table
- Other resource files (optional)

Darwin Core Data Package

DwC-DP table properties

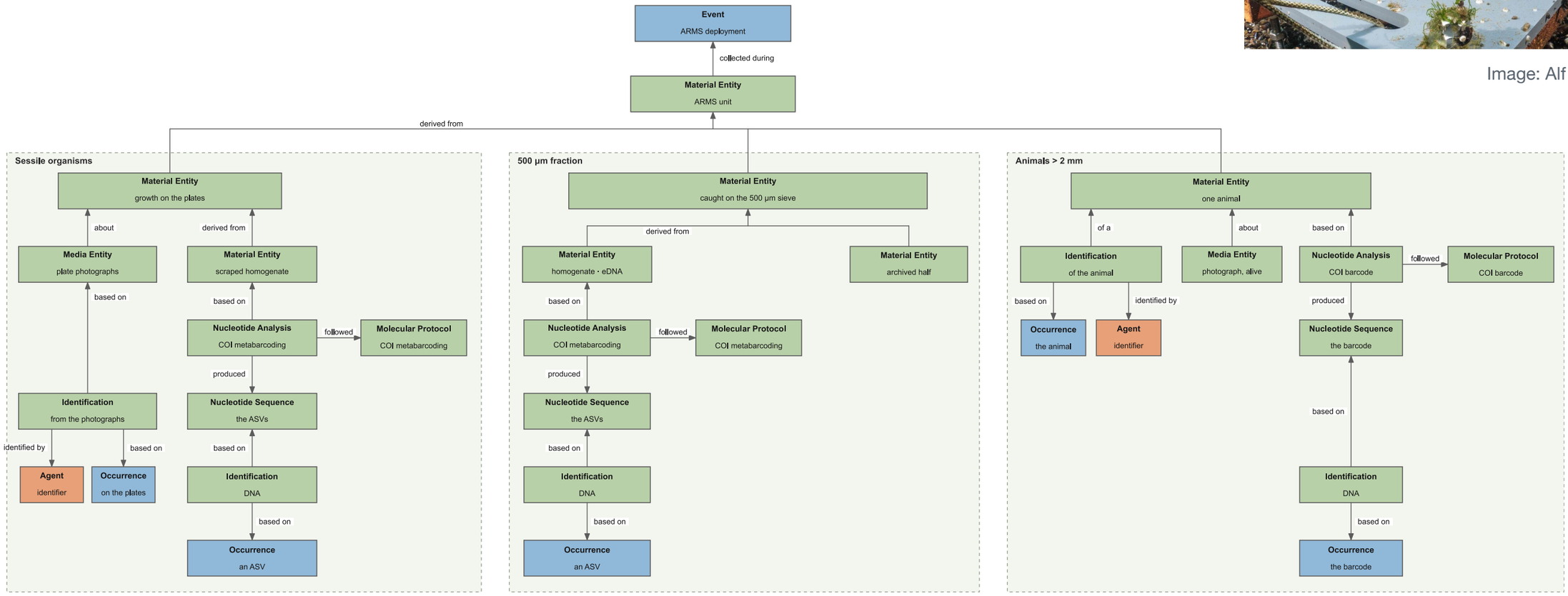
- Name
- Path
- Profile (must be `tabular-data-resource`)
- Format
- Media type (must be `text/csv`)
- Table schema (provided at `rs.tdwg.org`)
  - Fields
  - Primary key
  - Foreign keys (have a `predicate` property to document relationship semantics)

In practice

Example dataset: ARMS



Image: Alf Norkko



OBIS indexing

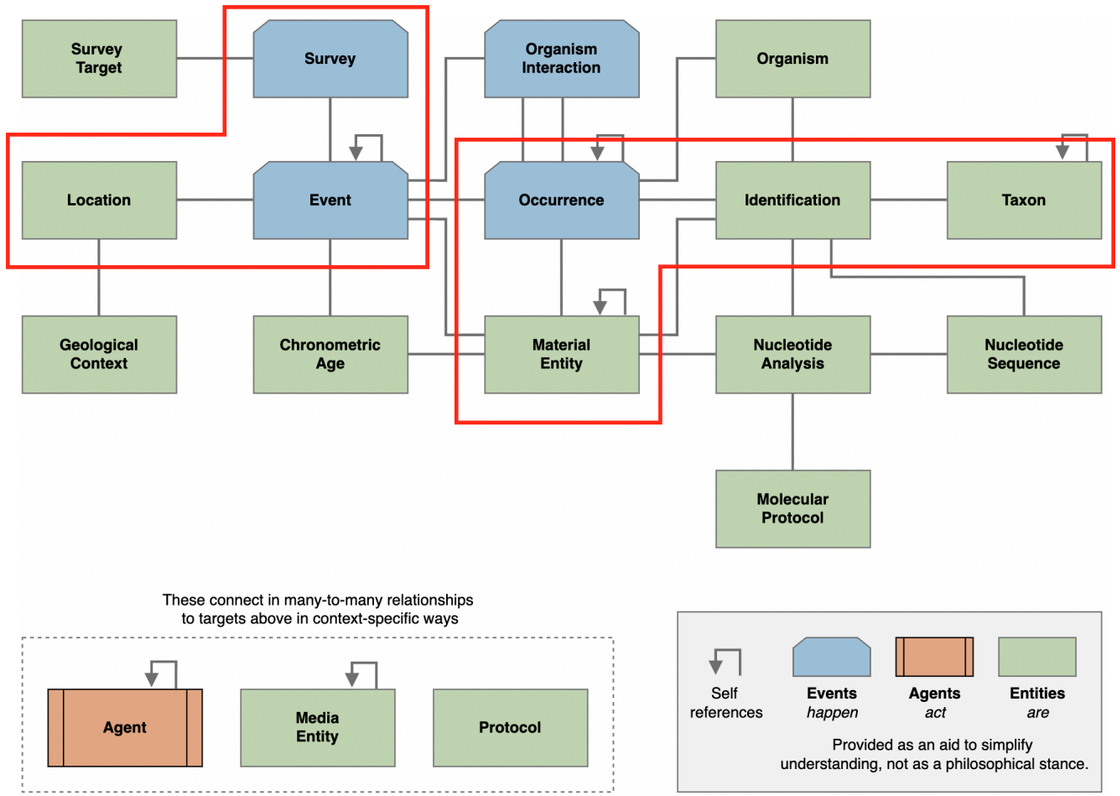
Implementation proposal

Phase 1

Project most common patterns to existing index (Event and Occurrence)

Phase 2

Provide SQL-like access to full data model



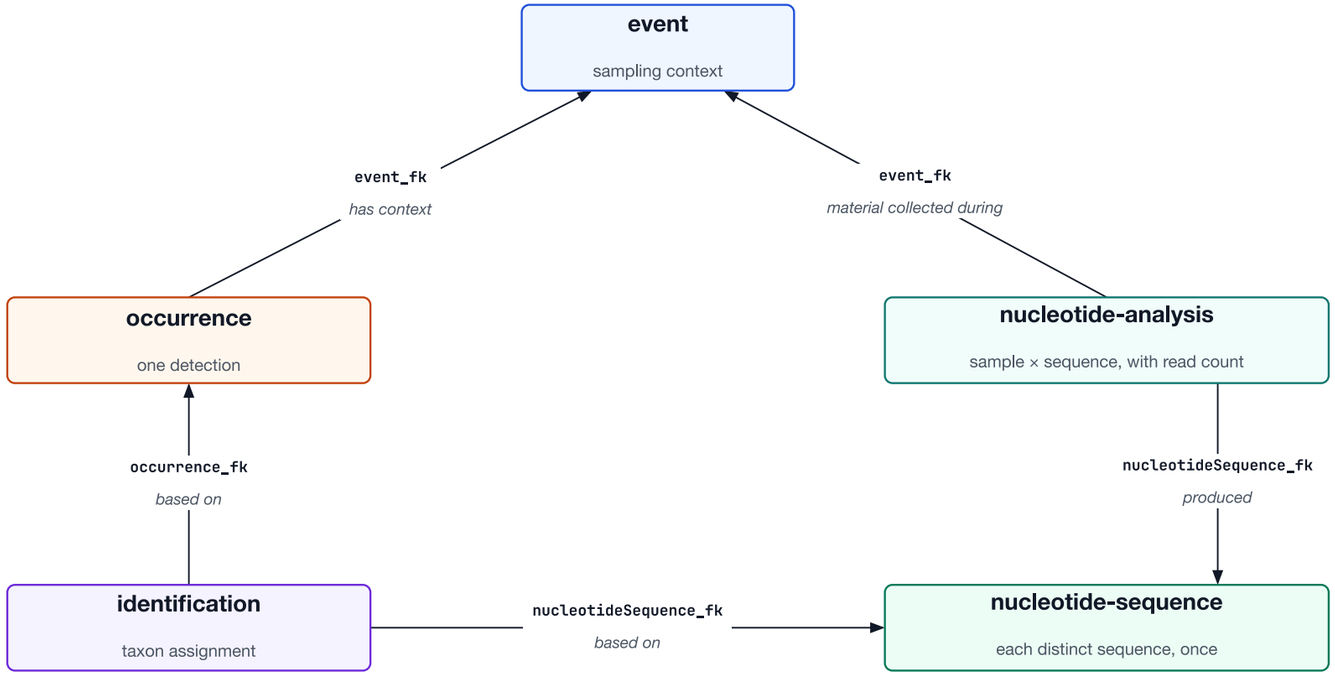
But… eDNA

The DNADerivedData extension is not a very efficient way of storing sequence data, as identical sequences are repeated for every sample.

```
➜  b508f7bc-8708-43d5-b940-e44f2765d1af ls -alh
total 319219120
drwx------  7 pieter  staff   224B Nov  4 12:16 .
drwxr-xr-x@ 5 pieter  staff   160B Nov  4 12:16 ..
-rw-rw-r--  1 pieter  staff    21G Dec 23  2025 dnaderiveddata.txt
-rw-rw-r--  1 pieter  staff   4.5K Dec 23  2025 eml.xml
-rw-rw-r--  1 pieter  staff   126G Dec 23  2025 extendedmeasurementorfact.txt
-rw-rw-r--  1 pieter  staff   5.0K Dec 23  2025 meta.xml
-rw-rw-r--  1 pieter  staff   5.9G Dec 23  2025 occurrence.txt
```

But… eDNA

DwC-DP offers a solution through the `nucleotide-analysis` table.



The other eDNA issue

Current indexing repeats Event level measurement at every occurrence, which is problematic in particular for high volume eDNA datasets.

